MILITARY SPECIFICATION

FILTER SET, GAS AND PARTICULATE, 200 CFM (MODULAR COLLECTIVE PROTECTION EQUIPMENT (MCPE)/SIMPLIFIED COLLECTIVE PROTECTION EQUIPMENT (SCPE) Applications)

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of cylindrical gas-particulate filter with an airflow rating of 200 cubic feet per minute (cfm).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-51525 - Filter, Gas, 200 CFM (MCPE/SCPE Applications).
MIL-F-51526 - Filter Particulate, 200 CFM (MCPE/SCPE Applications).

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 4240

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-19-6718 - Filter Set, Gas and Particulate.
5-19-6407 - Container, Filter.

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-19-6718 - Filter Set, Gas Particulate.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition; functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components. All materials and components of the filter set and package shall conform to the specific characteristics set forth on the drawings and specifications cited on Drawing 5-19-6718, Drawing 5-19-6407 and subsidiary drawings.

3.2 Preproduction. Prior to the start of regular production, a preproduction sample of packaged filter sets shall be produced in accordance with this specification for examination and test (see 4.3).

3.3 Workmanship. The filter set package shall be free of foreign matter (dirt, oil, or viscous material).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable-for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the , right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspeciton.

4.3.1 Lotting. A preproduction lot of 20 packaged filter sets shall be produced using the same methods, materials, equipment, and processes as will be used during regular production (see 6.2).

4.3.2 Sampling.

4.3.2.1 For examination. Each filter set unit pack in the preproduction lot shall be examined.

4.3.2.2 For packaging test. Each filter set (Drawing 5-19-6407) shall be tested.

4.3.2.3 For destructive test of gas filter. Sampling of packaged filter sets for destructive test of the gas filters shall be conducted in accordance with the preproduction inspection test requirements specified in MIL-F-51525 for destructive tests.

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4.3.3 Inspection procedure.

4.3.3.1 For examination. Each filter set package shall be examined for defects listed in the classification of defects (4.4.3.4). The AQLs listed in the classification of defects do not apply. If any packaged filter set is found to be defective, the preproduction lot shall be rejected.

4.3.3.2 For packaging test. Each filter set package (Drawing 5-19-6407) shall be tested for leakage in accordance with 4.4.4.1. If any filter con. tainer fails to meet the requirements for leakage (SPI P5-19-6718)) the preproduction lot shall be rejected.

4.3.3.3 For destructive test of gas filter. Sample gas filters from the packaged filter sets shall be inspected in accordance with the preproduction inspection, inspection procedures, for destructive tests, specified in MIL-F-51525.

4.3.4 Acceptance/rejection criteria. The preproduction lot shall comply with all requirements of this specification when examined and tested in accordance with 4.3 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the packaged filter sets of one type produced by one manufacturer, from the same materials, at the same location, under essentially the same manufacturing conditions, and at essentially the same time.

4.2.2 Sampling.

4.2.2.1 For examination. Sampling of packaged filter set shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For non-destructive tests. Each packaged filter set in the lot shall be tested.

4.4.2.3 For destructive tests. Packaged filter sets shall be taken at random from the lot in accordance with MIL-STD-414, level II, AQL 1.0 percent, and four additional packaged filter sets shall be taken at random from the lot to make up the total sample.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample packaged filter sets shall be examined for defects in accordance with MIL-STD-105.

4.4.3.2 For non-destructive tests. Each filter set package (Drawing 5-19-6407) shall be tested for leakage in accordance with 4.4.4.1. Any filter set failing to meet the requirements for leakage (SPI P5-19-6718) shall be rejected.

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4.4.3.3 For destructive tests. The following tests shall be performed sequentially on each packaged filter set in the sample.

(a) Leakage of packaged filter set. The sample of filter sets (4.4.2.3) shall be tested for the critical defect listed in the classification of defects (4.4.3.4(a)). If any sample packaged filter set fails the leakage requirements (SPI P5-19-6718), the lot shall be rejected.

(b) Unpackaged and disassembled filter set. The sample unpackaged and disassembled filter sets shall be examined for the defects in the list of defects (4.4.3.5). The gas filter and particulate filter shall be examined for the workmanship requirements specified in their respective specifications, MIL-F-51525 and MIL-F-51526. If a defect is found in either set, gas filter or particulate filter of any sample (4.4.2.3), the lot shall be rejected.

(c) Gas filter test. Four sample gas filters shall be tested for CK gas life in accordance with MIL-F-51525. The remaining samples shall be tested for airflow resistance and DMMP gas-life after rough handling in accordance with MIL-F-51525. The acceptance criteria of MIL-F-51525 shall apply for both tests.

4.4.3.4 Classification of defects.

(a) Packaged filter set (section 5) (Drawing 5-19-6407).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.4.4.1
Major:	None defined	
Minor:	AQL 2.5 percent defective	
201	Container closure incomplete (voids or missing cable)	
202	Marking missing, incorrect, or illegible	
203	Workmanship (3.3)	
	AQL 4.0 percent defective	
204	Dents or bulges more than l/4-inch	

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4.4.4 Test.

4.4.4.1 Package test. The filter set package (SPI P5-19-6718) shall be immersed in water so that its uppermost point is 0.5 feet maximum below the surface of the water. The air pressure shall be reduced to an absolute pressure of 6.2 + 0.2 pounds per square inch (12.6 inches + 0.4 inches of mercury) and the container examined for leakage.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in SPI P5-19-6718.

6. NOTES

6.1 Intended use. This specification covers a filter set intended for use with MCPE/SCPE components.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(c) Deliverable production filter sets. The filter sets not subjected to destructive testing may be considered deliverable items provided that the preproduction lot is accepted. Appropriate previsions should be made to deliver these items to the designated activity.
(d) Shelf life code.
(e) Level of packing (see 6.3 and SPI P5-19-6718).

6.3 Level C packing. Level C packing as defined in SPI P5-19-6718 is for shipments of filter sets from a vendor or supplier to a prime contractor, or from a prime contractor to a military installation or domestic destination for immediate use, or for further processing.

6.4 Subject term (key word) listing.

Filter, gas
Filter, particulate
Filter set, gas-particulate
Filter set, packaged

6.5 Supersession data. This specification supersedes Chemical Systems Laboratory Purchase Description EA-F-1180A, dated 23 June 1980.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-51527(EA)	FILTER SET, GAS AND PARTICULATE, 200 CFM (MCPE/SCPE APPLICATIONS)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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